ACOMAN

Weather Display

Smart weather station & digital photo frame

User Guide · Firmware v0.6

Contents

1. What the device can do

Acoman Weather Display is a home weather station with a color touchscreen that shows current weather, forecasts, family photos, and can be fully controlled through a web browser – no app installation required.

5-day forecast

Hourly forecast

Custom photos

Main features

- **Current weather** – temperature, humidity, wind, UV index, cloud cover, sunset time
- **Hourly and 5-day forecast**
- **Digital photo frame** – custom photos on SD card, auto-resized to 240×320 px
- **RGB LED backlight** – static color or smooth rainbow fade
- **8 interface languages** – EN, ES, DE, FR, TR, SV, IT, SK
- **OTA firmware update** – upload new firmware without USB cable
- **Night mode** – automatic brightness dimming at night
- **Free weather data** – powered by Open-Meteo API, no API key needed

> **Tip**
> All settings are configured from your phone, tablet, or computer using a regular web browser – no app installation needed.

2. First installation (Flash Download Tool)

On a brand new device, you need to flash the firmware once via USB using the ESP32 Flash Download Tool. After this, all future updates can be done wirelessly (OTA).

Requirements

- USB cable
- [ESP32 Flash Download Tool (latest version)] (free, Windows)
- 4 firmware files from the **Releases** section on GitHub

Flash addresses

File	Address
`bootloader.bin`	`0x1000`
`partitions.bin`	`0x8000`
`boot_app0.bin`	`0xE000`
`acoman-weather-v0.6.bin`	`0x10000`

Flash Download Tool settings

Setting	Value
ChipType	`ESP32`
WorkMode	`Develop`
SPI Speed	`80 MHz`
SPI Mode	`DIO`
Baud	`921600`

Load all 4 files, set the correct addresses, and click **START**. Wait for **FINISH**, then disconnect and reconnect USB.

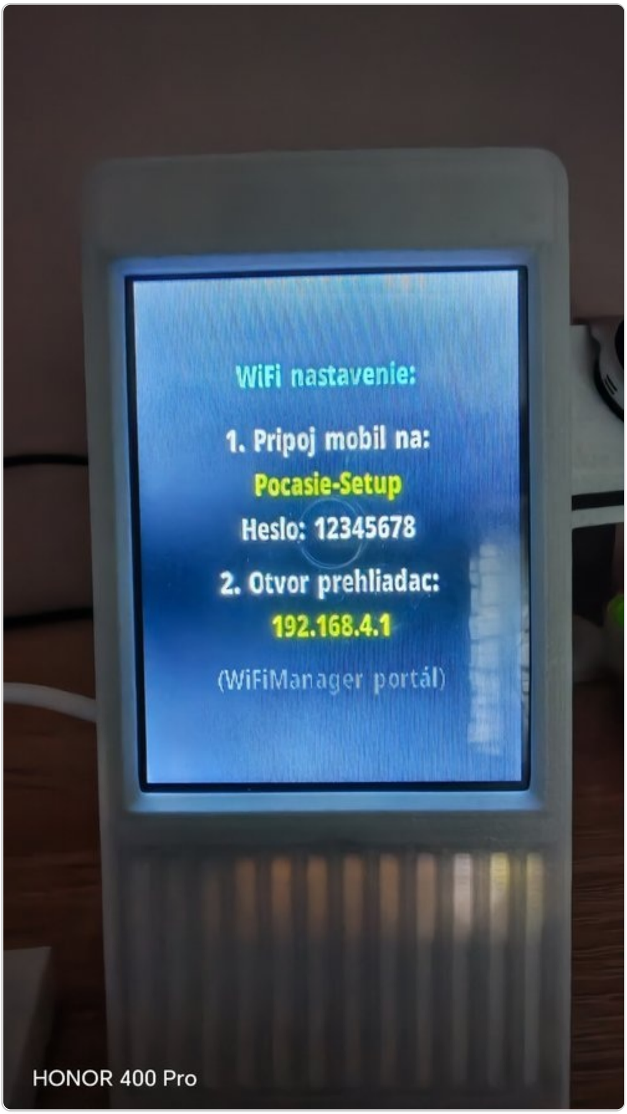
After flashing — device is ready

Important

This step is only required once for a brand new device. All future firmware updates are done wirelessly via the built-in OTA web interface.

3. First start and WiFi connection

After flashing, the device creates a temporary WiFi network for initial setup.



WiFi setup screen on first boot

1 Connect your phone

Connect to WiFi network `Acoman-Setup`
Password: `12345678`

2 Open your browser

The setup page usually opens automatically. If not, enter `192.168.4.1` in your browser.

3 Select your home WiFi

Choose your network, enter the password, and confirm. The device restarts and connects.

> **Note**
> This setup screen only appears on a brand new device or after pressing "Reset WiFi" in settings.

4. Web settings interface

After connecting to your home WiFi, open the settings page by long-pressing the display – the IP address will appear on screen.

Long press to show IP address

How to find the IP address

Press and hold the display (1.5 s) on the main weather screen – the IP address appears for entering in your browser.

The web interface contains: Language, Location, Photos, Display settings, RGB LED, WiFi and Firmware.

Open `http://[IP address]` in any browser on the same WiFi network.

5. Location and language

Changing the language

Select a language from the dropdown (8 languages available) and click **Save language**. The device restarts and the entire interface – web page and display – switches to the selected language.

Setting the weather location

Location settings and city search

Enter a city name in the **Search city** field and click **Search**. Click the correct result – coordinates (LAT/LON) fill in automatically.

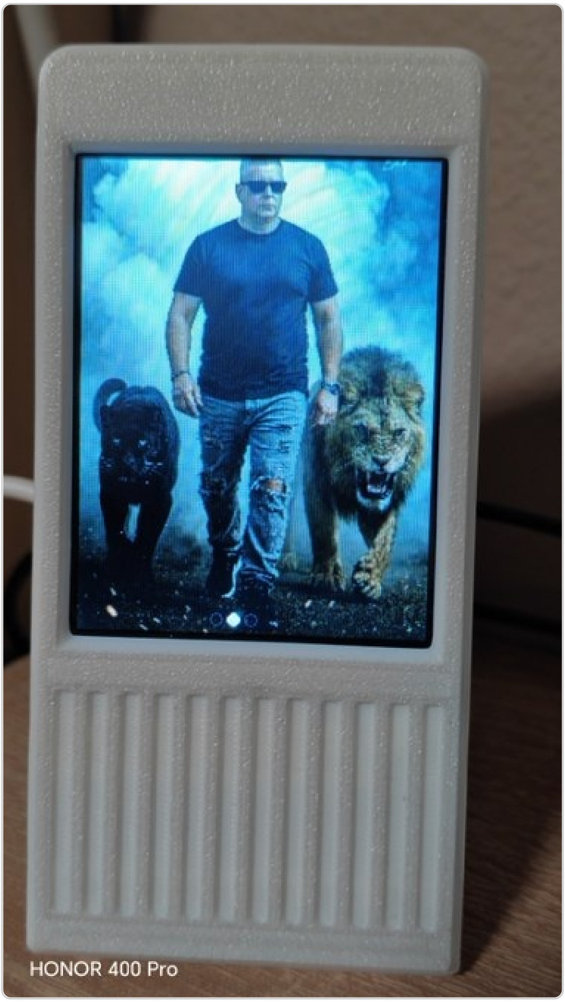
The city name shown on the display can be edited in the **City name** field.

Optionally enable **°F (Fahrenheit)** – temperature will switch to °F and wind speed to mph.

Click **Save and restart** to apply and download new weather data.

6. Photos on display (SD card)

The device supports displaying custom photos as a digital photo frame. Photos are automatically resized to 240×320 px.



Custom photo on display

Storage	Capacity
Internal (SPIFFS)	~128–700 KB
SD card (microSD)	depends on card

Adding a photo

1. Click **Choose file** and select a photo
2. Photo is auto-cropped to 240×320 px, preview shown
3. Click **Upload photo to display**

Each photo has a **Delete** button to remove it individually.

Tip

For more photos, insert a microSD card – the slot is on the board.

7. Display brightness and night mode

In the **Display settings** section you can:

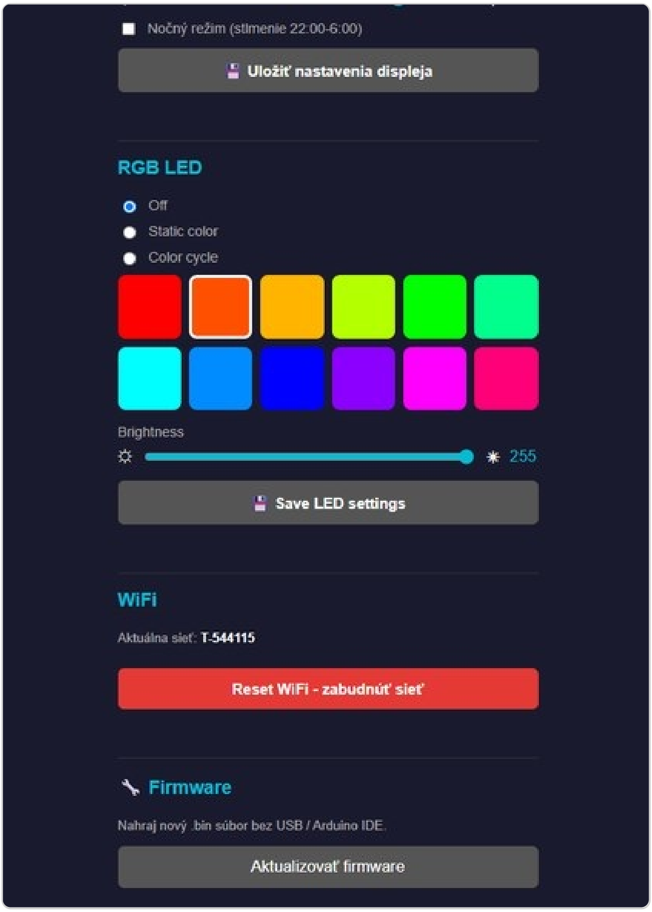
- Set **display brightness** with a slider (10–255)
- Enable **night mode** – automatically dims the display between 10 PM and 6 AM

Click **Save display settings**. Changes take effect immediately, no restart needed.

8. RGB LED lighting

The device includes a color RGB LED with three modes:

Mode	Description
Off	LED is turned off
Static color	One fixed color from 12 preset shades
Color cycle	Smooth rainbow fade through the full color spectrum



12-color palette and settings

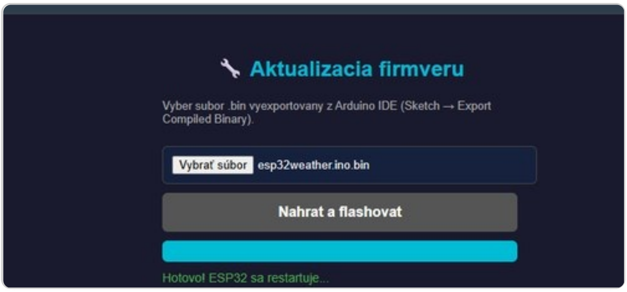
RGB LED ambient lighting in action

- **Brightness** – overall LED brightness (5–255)
- **Cycle speed** – time for one full spectrum pass (0.5–15 seconds)

Click **Save LED settings** to save permanently.

9. Firmware update (OTA without USB)

After the first installation, all future updates are done wirelessly – no USB cable or Arduino IDE needed.



Firmware update page

1 Open Firmware page

In settings, click **Update firmware** at the bottom of the page.

2 Select .bin file

Download the latest `acoman-weather-vX.X.bin` from GitHub Releases and select it.

3 Upload and flash

Click **Upload and flash**, confirm the dialog. Progress bar shows upload status. Device restarts automatically.

> **Important**
> Do not disconnect power during firmware upload. Interrupting may require USB recovery.

10. Touch controls on display

Gesture	Action
Tap left edge	Previous screen
Tap right edge	Next screen
Swipe left / right	Change screen
Tap center	Pause auto-rotation for 60 seconds
Long press (1.5 s)	Show IP address for settings

The device automatically cycles through screens (current weather, hourly forecast, 5-day forecast, photos) every 10 seconds.

11. Troubleshooting

Device does not connect to WiFi

Long press the display, go to settings and click **Reset WiFi – forget network**. The WiFi setup screen appears again (see chapter 3).

Photos are not showing

- Check that the SD card is correctly inserted (FAT32 format)
- Check free space in the "Photos on display" section
- Try uploading the photo again

Settings page does not load

- Make sure your phone/PC is on the same WiFi network as the device
- Long press the display to check the current IP address – it may change after router restart

Firmware update failed

- Make sure you selected the correct `.bin` file
- Try again with a stable WiFi signal

12. Technical specifications and pins

Hardware

Component	Description
Microcontroller	ESP32-WROOM-32E
Display	2.8″ TFT 240×320 px, ST7789 driver
Touch	Resistive (RTP), XPT2046
Connectivity	WiFi 2.4 GHz
Photo storage	Internal SPIFFS or microSD card
RGB LED	Common anode, 3-channel (GPIO 16, 17, 22)
SKU	E32R28T-1

Key pin wiring

SD card – CS	GPIO 5
SD card – SCK	GPIO 18
SD card – MISO	GPIO 19
SD card – MOSI	GPIO 23
LCD backlight	GPIO 21
RGB LED – R	GPIO 17
RGB LED – G	GPIO 22
RGB LED – B	GPIO 16

Default credentials

Purpose	Value
WiFi setup network (SSID)	`Acoman-Setup`
WiFi setup password	`12345678`
OTA hostname	`Acoman`